

Calculus I Lesson 13

Formulas: $(a - b)(a + b) = a^2 - b^2$ and $(\sqrt{a + b})^2 = a + b$

1) $(\sqrt{x + 2} - \sqrt{x}) \cdot (\sqrt{x + 2} + \sqrt{x}) = \underline{\hspace{2cm}} ?$

2) $(\sqrt{x - 1} - \sqrt{x}) \cdot (\sqrt{x - 1} + \sqrt{x}) = \underline{\hspace{2cm}} ?$

$$3) \frac{\sqrt{x-2} - \sqrt{x}}{2} = \frac{(\sqrt{x-2} - \sqrt{x})}{2} \cdot \frac{(\sqrt{x-2} + \sqrt{x})}{(\sqrt{x-2} + \sqrt{x})} = \underline{\hspace{2cm} ? \hspace{2cm}}$$

$$4) \frac{\sqrt{x+4} - \sqrt{x}}{4} = \frac{(\sqrt{x+4} - \sqrt{x})}{4} \cdot \frac{(\sqrt{x+4} + \sqrt{x})}{(\sqrt{x+4} + \sqrt{x})} = \underline{\hspace{2cm} ? \hspace{2cm}}$$

$$5) \frac{\sqrt{x+h} - \sqrt{x}}{h} = \frac{(\sqrt{x+h} - \sqrt{x})}{h} \cdot \frac{(\sqrt{x+h} + \sqrt{x})}{(\sqrt{x+h} + \sqrt{x})} = \underline{\hspace{2cm} ? \hspace{2cm}}$$

$$6) \frac{\sqrt{x+\Delta x} - \sqrt{x}}{\Delta x} = \frac{(\sqrt{x+\Delta x} - \sqrt{x})}{\Delta x} \cdot \frac{(\sqrt{x+\Delta x} + \sqrt{x})}{(\sqrt{x+\Delta x} + \sqrt{x})} = \underline{\hspace{2cm} ? \hspace{2cm}}$$

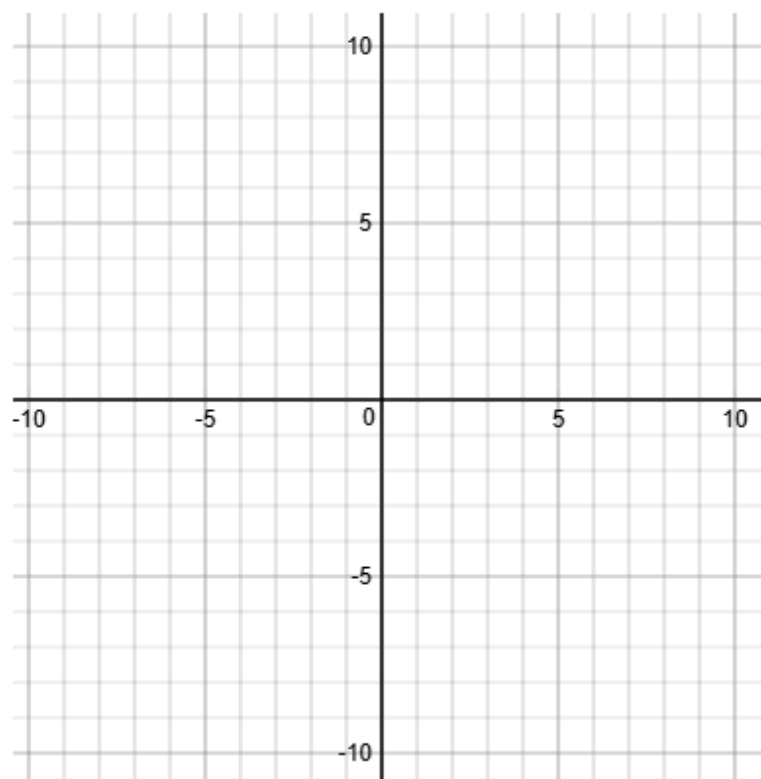
7) $f(x) = \sqrt{x}$

a) Find the difference quotient: $\frac{f(x + \Delta x) - f(x)}{\Delta x}$

b) $f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = \underline{\hspace{2cm}} ?$

c) $f(4) = \underline{\hspace{2cm}} ?$ and $f'(4) = \underline{\hspace{2cm}} ?$

d) Graph $f(x) = \sqrt{x}$ and the line $y = f'(4) \cdot (x - 4) + 2$



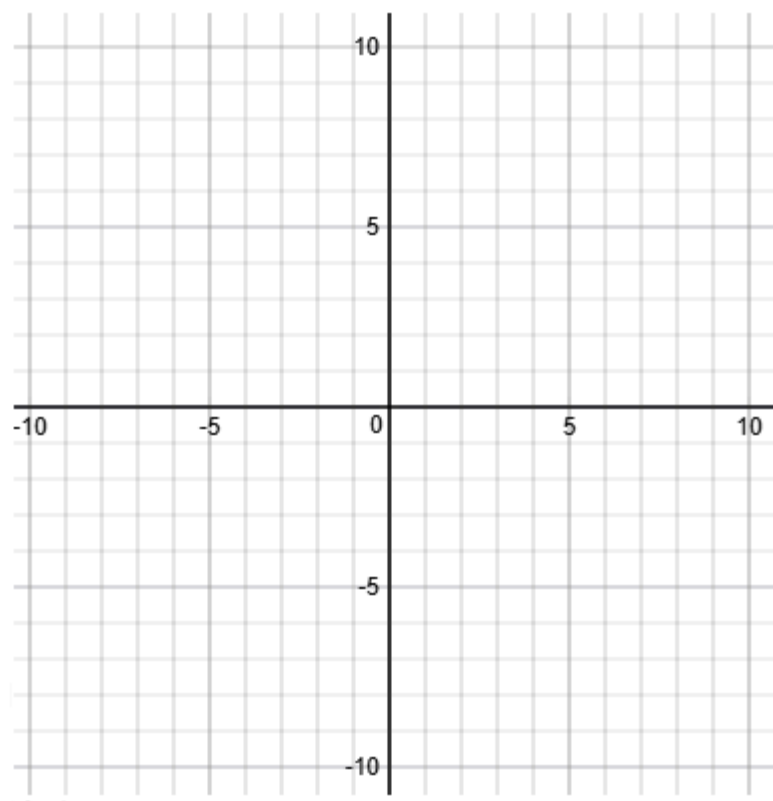
8) $f(x) = \sqrt{x+2}$

a) Find the difference quotient: $\frac{f(x + \Delta x) - f(x)}{\Delta x}$

b) $f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = \underline{\hspace{1cm}} ?$

c) $f(2) = \underline{\hspace{1cm}} ?$ and $f'(2) = \underline{\hspace{1cm}} ?$

d) Graph $f(x) = \sqrt{x+2}$ and
the line $y = f'(2) \cdot (x - 2) + 2$



9) $f(x) = \sqrt{2x+1}$

Hint: $f(x + \Delta x) = \sqrt{2(x + \Delta x) + 1} = \sqrt{2x + 2\Delta x + 1}$

a) Find the difference quotient: $\frac{f(x + \Delta x) - f(x)}{\Delta x}$

b) $f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = \underline{\hspace{2cm}} ?$

c) $f(4) = \underline{\hspace{2cm}} ?$ and $f'(4) = \underline{\hspace{2cm}} ?$

d) Graph $f(x) = \sqrt{2x+1}$ and
the line $y = f'(4) \cdot (x - 4) + 3$

